Task 04: Visual Reconnaissance & Live Asset Identification

In this task, we filtered out inactive subdomains and focused only on live web assets. Tools like **Httprobe** and **Httpx** were used to verify which subdomains are active.

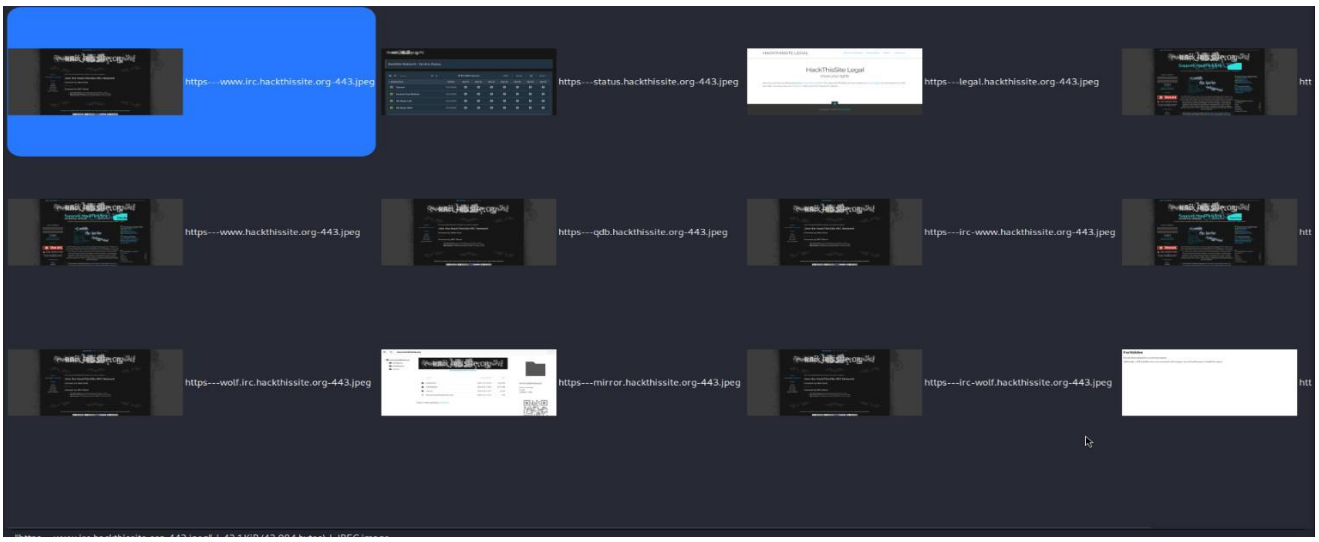
```
┌──(hyena㉿hyena)-[~/Desktop/Cybyster/Week01/Task_01]
└─$ cat all_subdomain.txt | httpx -silent > live_subdomains.txt

┌──(hyena㉿hyena)-[~/Desktop/Cybyster/Week01/Task_01]
└─$ cat all_subdomain.txt
 (ASN) --> announces --> 2406:d500:7::/48 (Netblock)
 (ASN) --> announces --> 5.220.0.0/14 (Netblock)
 (ASN) --> announces --> 89.106.200.0/21 (Netblock)
 (ASN) --> managed_by --> Not routed (RIROrganization)
-v3dev-cdn.hackthissite.org (FQDN) --> cname_record --> v3dev.hackthissite.org (FQDN)
-v3stage-cdn.hackthissite.org (FQDN) --> cname_record --> v3stage-cdn.hackthissite.org (FQDN)
01.187.74.137.in-addr.arpa (FQDN) --> ptr_record --> hackthissite.org (FQDN)
02.187.74.137.in-addr.arpa (FQDN) --> ptr_record --> hackthissite.org (FQDN)
03.25.56.0/22 (Netblock) --> contains --> 103.25.56.55 (IPAddress)
15.187.74.137.in-addr.arpa (FQDN) --> ptr_record --> api.hackthissite.org (FQDN)
16.202.0.0/15 (Netblock) --> contains --> 116.203.6.3 (IPAddress)
17.187.74.137.in-addr.arpa (FQDN) --> ptr_record --> api.hackthissite.org (FQDN)
18.187.74.137.in-addr.arpa (FQDN) --> ptr_record --> api.hackthissite.org (FQDN)
19.187.74.137.in-addr.arpa (FQDN) --> ptr_record --> api.hackthissite.org (FQDN)
23.187.74.137.in-addr.arpa (FQDN) --> ptr_record --> pi.hackthissite.org (FQDN)
33.187.74.137.in-addr.arpa (FQDN) --> ptr_record --> mirror.hackthissite.org (FQDN)
36557 (ASN) --> announces --> 103.25.56.0/22 (Netblock)
36557 (ASN) --> announces --> 2406:d500:2::/48 (Netblock)
36557 (ASN) --> managed_by --> HOST-AS-AP Host Universal Pty Ltd (RIROrganization)
37.74.0.0/16 (Netblock) --> contains --> 137.74.187.100 (IPAddress)
37.74.0.0/16 (Netblock) --> contains --> 137.74.187.101 (IPAddress)
37.74.0.0/16 (Netblock) --> contains --> 137.74.187.102 (IPAddress)
37.74.0.0/16 (Netblock) --> contains --> 137.74.187.103 (IPAddress)
37.74.0.0/16 (Netblock) --> contains --> 137.74.187.104 (IPAddress)
37.74.0.0/16 (Netblock) --> contains --> 137.74.187.115 (IPAddress)
37.74.0.0/16 (Netblock) --> contains --> 137.74.187.116 (IPAddress)
37.74.0.0/16 (Netblock) --> contains --> 137.74.187.117 (IPAddress)
37.74.0.0/16 (Netblock) --> contains --> 137.74.187.118 (IPAddress)
37.74.0.0/16 (Netblock) --> contains --> 137.74.187.119 (IPAddress)
37.74.0.0/16 (Netblock) --> contains --> 137.74.187.133 (IPAddress)
37.74.0.0/16 (Netblock) --> contains --> 137.74.187.134 (IPAddress)
37.74.0.0/16 (Netblock) --> contains --> 137.74.187.135 (IPAddress)
37.74.0.0/16 (Netblock) --> contains --> 137.74.187.136 (IPAddress)
37.74.0.0/16 (Netblock) --> contains --> 137.74.187.137 (IPAddress)
37.74.0.0/16 (Netblock) --> contains --> 137.74.187.138 (IPAddress)
37.74.0.0/16 (Netblock) --> contains --> 137.74.187.145 (IPAddress)
37.74.0.0/16 (Netblock) --> contains --> 137.74.187.146 (IPAddress)
37.74.0.0/16 (Netblock) --> contains --> 137.74.187.150 (IPAddress)
37.74.0.0/16 (Netblock) --> contains --> 137.74.187.151 (IPAddress)
37.74.0.0/16 (Netblock) --> contains --> 137.74.187.152 (IPAddress)
37.74.0.0/16 (Netblock) --> contains --> 137.74.187.153 (IPAddress)
37.74.0.0/16 (Netblock) --> contains --> 137.74.187.154 (IPAddress)
37.74.0.0/16 (Netblock) --> contains --> 137.74.187.155 (IPAddress)
37.74.0.0/16 (Netblock) --> contains --> 137.74.187.157 (IPAddress)
37.74.0.0/16 (Netblock) --> contains --> 137.74.187.158 (IPAddress)
37.74.0.0/16 (Netblock) --> contains --> 137.74.187.159 (IPAddress)
42.250.146.0/23 (Netblock) --> contains --> 142.250.147.27 (IPAddress)
42.251.127.0/24 (Netblock) --> contains --> 142.251.127.27 (IPAddress)
46.187.74.137.in-addr.arpa (FQDN) --> ptr_record --> staff.hackthissite.org (FQDN)
50.187.74.137.in-addr.arpa (FQDN) --> ptr_record --> lille.irc.hackthissite.org (FQDN)
51.187.74.137.in-addr.arpa (FQDN) --> ptr_record --> lille.irc.hackthissite.org (FQDN)
5169 (ASN) --> announces --> 142.250.146.0/23 (Netblock)
5169 (ASN) --> announces --> 142.251.127.0/24 (Netblock)
5169 (ASN) --> announces --> 172.217.221.0/24 (Netblock)
5169 (ASN) --> announces --> 172.253.0.0/16 (Netblock)
5169 (ASN) --> announces --> 192.178.0.0/15 (Netblock)
```

Aquatone and **Gowitness** captured screenshots of these live assets. This process helps identify valuable targets such as login panels or development environments, allowing us to prioritize and save time during reconnaissance.

Live domain being visited

```
┌──(hyena㉿hyena)-[~/Desktop/Cybyster/Week01/Task_01]
└─$ gowitness scan file  -f live_subdomains.txt
026/03/07 23:25:33 WARN no writers have been configured. to persist probe results, add writers using --write-* flags
026/03/07 23:25:40 INFO result 🤖 target=https://wolf.irc.hackthissite.org:443 status-code=200 title="HackThisSite IRC" have-screenshot=true
026/03/07 23:25:40 INFO result 🤖 target=https://www.irc.hackthissite.org:443 status-code=200 title="HackThisSite IRC" have-screenshot=true
026/03/07 23:25:47 INFO result 🤖 target=https://irc-www.hackthissite.org:443 status-code=200 title="HackThisSite IRC" have-screenshot=true
026/03/07 23:25:56 INFO result 🤖 target=https://status.hackthissite.org:443 status-code=200 title="HackSite Network - Service Status - Powered by HetrixTools" have-screen
026/03/07 23:26:02 INFO result 🤖 target=https://mirror.hackthissite.org:443 status-code=200 title=mirror.hackthissite.org have-screenshot=true
026/03/07 23:26:09 INFO result 🤖 target=https://legal.hackthissite.org:443 status-code=200 title="HackThisSite Legal" have-screenshot=true
026/03/07 23:26:14 INFO result 🤖 target=https://qdb.hackthissite.org:443 status-code=200 title="HackThisSite IRC" have-screenshot=true
026/03/07 23:26:21 INFO result 🤖 target=https://irc-wolf.hackthissite.org:443 status-code=200 title="HackThisSite IRC" have-screenshot=true
026/03/07 23:26:28 INFO result 🤖 target=https://hackthissite.org:443 status-code=200 title="Hack This Site" have-screenshot=true
026/03/07 23:26:32 INFO result 🤖 target=https://www.hackthissite.org:443 status-code=200 title="Hack This Site" have-screenshot=true
026/03/07 23:26:33 INFO result 🤖 target=https://ctf.hackthissite.org:443 status-code=200 title="35C3 Junior CTF Write-ups Posted | HackThisSite CTF" have-screenshot=false
026/03/07 23:26:33 INFO result 🤖 target=https://irc.hackthissite.org:443 status-code=200 title="HackThisSite IRC" have-screenshot=false
026/03/07 23:26:33 INFO result 🤖 target=https://h5ai.hackthissite.org:443 status-code=200 title="HackThisSite / h5ai" have-screenshot=false
026/03/07 23:26:34 INFO result 🤖 target=https://lille.irc.hackthissite.org:443 status-code=200 title="HackThisSite IRC" have-screenshot=false
026/03/07 23:26:39 INFO result 🤖 target=https://forums.hackthissite.org:443 status-code=403 title="403 Forbidden" have-screenshot=true
026/03/07 23:26:40 INFO result 🤖 target=https://hp.hackthissite.org:443 status-code=200 title="Hack This Site" have-screenshot=true
026/03/07 23:26:40 INFO result 🤖 target=https://stats.hackthissite.org:443 status-code=200 title="" have-screenshot=false

┌──(hyena㉿hyena)-[~/Desktop/Cybyster/Week01/Task_01]
└─$ ls
all_subdomain.txt    assetfinder.png    Combining_Reult.png   dnsdumbetser.png   hackthissite_subdomains.txt   live_subdomains.txt   screenshots        Subfinder.png
ammas-scanning.png   assetfinder.txt    crt.sh.png            'final script'     httpx.png                     output.txt            'sh terminal.png'   subfinder.txt

┌──(hyena㉿hyena)-[~/Desktop/Cybyster/Week01/Task_01]
└─$ cd screenshots

┌──(hyena㉿hyena)-[~/…/Cybyster/Week01/Task_01/screenshots]
└─$ ls
ttps---forums.hackthissite.org-443.jpeg  https---irc-wolf.hackthissite.org-443.jpeg  https---mirror.hackthissite.org-443.jpeg  https---wolf.irc.hackthissite.org-443.jpeg
ttps---hackthissite.org-443.jpeg         https---irc-www.hackthissite.org-443.jpeg   https---qdb.hackthissite.org-443.jpeg     https---www.hackthissite.org-443.jpeg
ttps---hp.hackthissite.org-443.jpeg      https---legal.hackthissite.org-443.jpeg     https---status.hackthissite.org-443.jpeg  https---www.irc.hackthissite.org-443.jpeg

┌──(hyena㉿hyena)-[~/…/Cybyster/Week01/Task_01/screenshots]
└─$
```

Screen shot captured by tool



Weekly Deliverable: The Reconnaissance Report– Summary

1 Asset List

Main Subdomains & IPs

Subdomain	IP Address(es)	Record Type	Status
hackthissite.org	137.74.187.100-104	A	Live
www.hackthissite.org	137.74.187.100-104	A	Live
api.hackthissite.org	137.74.187.115-119	A	Live
irc.hackthissite.org	137.74.187.150, 185.24.222.13	A	Live
mirror.hackthissite.org	137.74.187.133-134	A	Live
git.hackthissite.org	137.74.187.145	A	Live
ctf.hackthissite.org	-	CNAME (GitHub Pages)	Live
h5ai.hackthissite.org	-	CNAME (GitHub Pages)	Live
v3dev.hackthissite.org	198.148.81.145	A	Discovered
v3stage.hackthissite.org	198.148.81.147	A	Discovered

Infrastructure IP Ranges

Netblock	ASN	Provider	Location
137.74.0.0/16	16276	OVH	France
185.24.220.0/22	196752	TILAA	-
198.148.80.0/20	46844	Sharktech	-
103.25.56.0/22	136557	Host Universal Pty Ltd	-

Netblock	ASN	Provider	Location
142.251.127.0/24	15169	Google LLC	US

Mail Servers

Mail Server	Priority	IP
alt1.aspmx.l.google.com	-	192.178.213.27
aspmx.l.google.com	-	142.251.127.27

Name Servers

NS	IP
c.ns.buddyns.com	116.203.6.3
f.ns.buddyns.com	5.223.55.119

2 Technology Map

Web Technologies & Services

Subdomain	Technologies	Status
hackthissite.org	jQuery 1.8.1, Varnish	200
www.hackthissite.org	jQuery, Varnish	200
h5ai.hackthissite.org	GitHub Pages	200
ctf.hackthissite.org	GitHub Pages, Varnish	-
api.hackthissite.org	Load-balanced API	200
git.hackthissite.org	Git repository	200
forums.hackthissite.org	-	403

Services & Infrastructure

Web Servers: Varnish, GitHub Pages

Mail: Google Workspace

IRC Servers: Multiple (irc, irc-www, irc-wolf, wolf.irc)

Development/Staging: v3dev, v3stage

CDN: v3dev-cdn, v3stage-cdn

Monitoring: stats.hackthissite.org

File Sharing: h5ai.hackthissite.org

Staff Portal: staff.hackthissite.org

Load Balancing: Multiple A records for main & API domains

Geographic Distribution: France, US, UK, Australia

IPv6 Support: Enabled on multiple subdomains

3 Security Gaps (Low-Hanging Fruit)

Finding	Risk / Impact
Exposed .git directory	High – Source code, credentials, API keys can be leaked
Development/Staging subdomains	Medium – Weak security, debug info, test data exposure
Subdomain takeover potential (GitHub Pages, redirect.pizza)	Medium – Service impersonation, phishing
Legacy / historical subdomains from SSL certificates	Medium – Could be re-registered, info leakage
Infrastructure hosts exposed (vm-XXX.firewall)	Low – Reveals internal naming & architecture patterns
PTR records reveal IP/service mapping	Low – Infrastructure info leakage
Outdated jQuery 1.8.1	Low – Known vulnerabilities
Google Workspace mail server info	Info – Reveals email provider & naming convention

Finding	Risk / Impact
IPv6 addresses exposed	Info – Expanded attack surface

Summary of Critical Findings

High: Exposed .git – full code & credential leakage

Medium: Development/staging subdomains & subdomain takeover potential

Low/Info: Infrastructure naming, outdated tech, multiple load-balanced IPs, IPv6 enabled